Check point 11 paragraphe :

The primary goal of optimization in training a neural network is to find the optimal set of weights and biases that minimizes the loss function, which measures the difference between the model's predictions and the actual outcomes. Optimizers like Stochastic Gradient Descent (SGD) or Adam achieve this by calculating the gradients of the loss and iteratively updating the model's parameters in the direction that reduces the overall error. This iterative process is heavily influenced by hyperparameters, where the learning rate controls the size of the parameter updates, the batch size dictates how many data samples are processed before an update occurs, and the number of epochs defines how many full passes the model makes through the entire training dataset. Carefully tuning these hyperparameters is crucial for achieving good generalization, as it prevents the model from either failing to learn the data's patterns (underfitting) or memorizing the training data along with its noise (overfitting). For example, if the learning rate is too high, the model may take steps that are too large, overshooting the optimal solution and causing the training process to diverge rather than converge on the lowest loss